JIONEX
TECHNOLOGY EXPERTISE

01

Voter Registration (Pre-Election)

NRBs register at embassies or via the BD PASS NRB portal

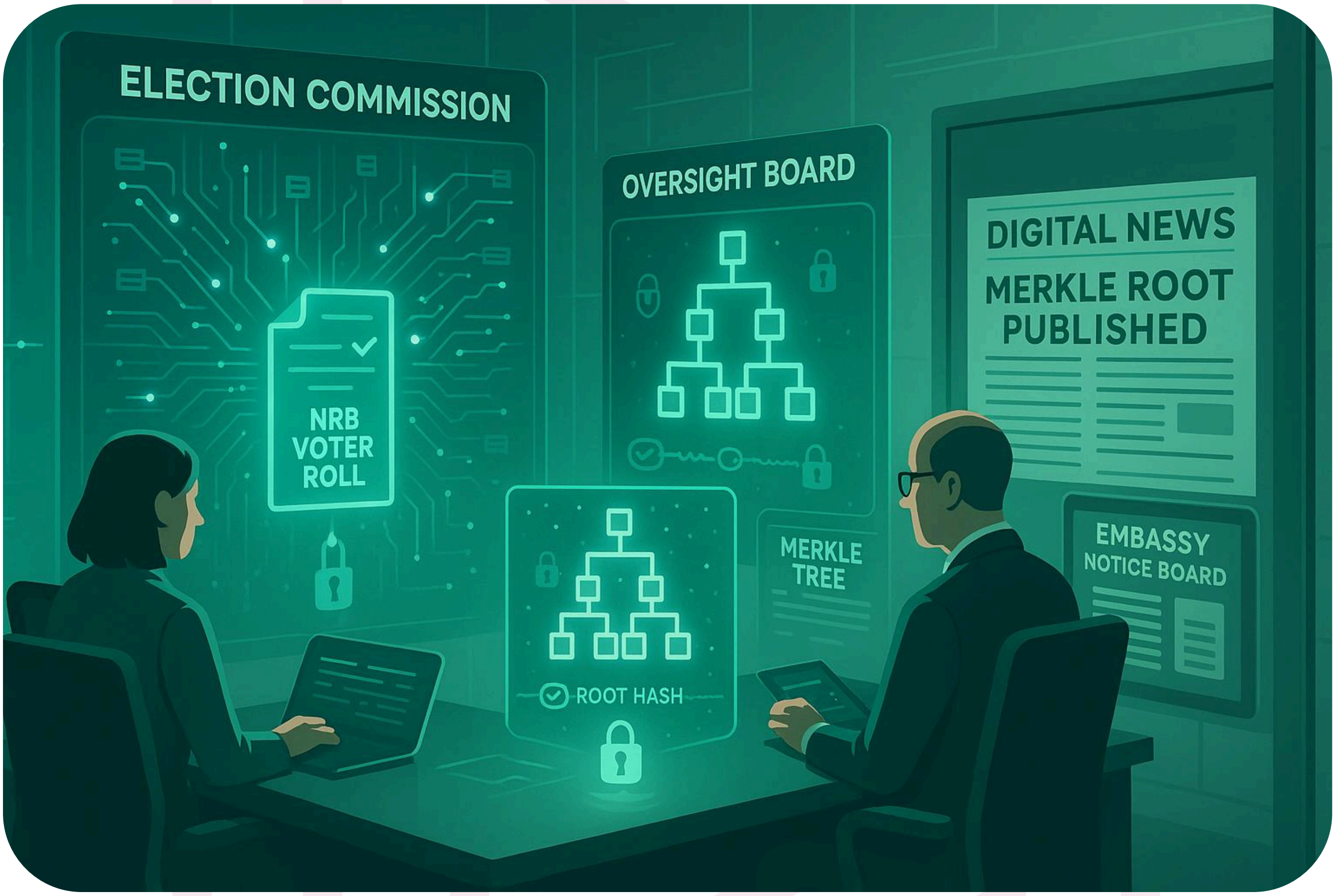
Present NID + passport + visa/residency

Voters are issued:

One-Time Voting Token (OTVT) — unlinkable to identity

ZK membership credential (used for proof-of-eligibility)

Final Voter Roll Generation



EC + Oversight Board (multi-party computation) compile final NRB voter roll

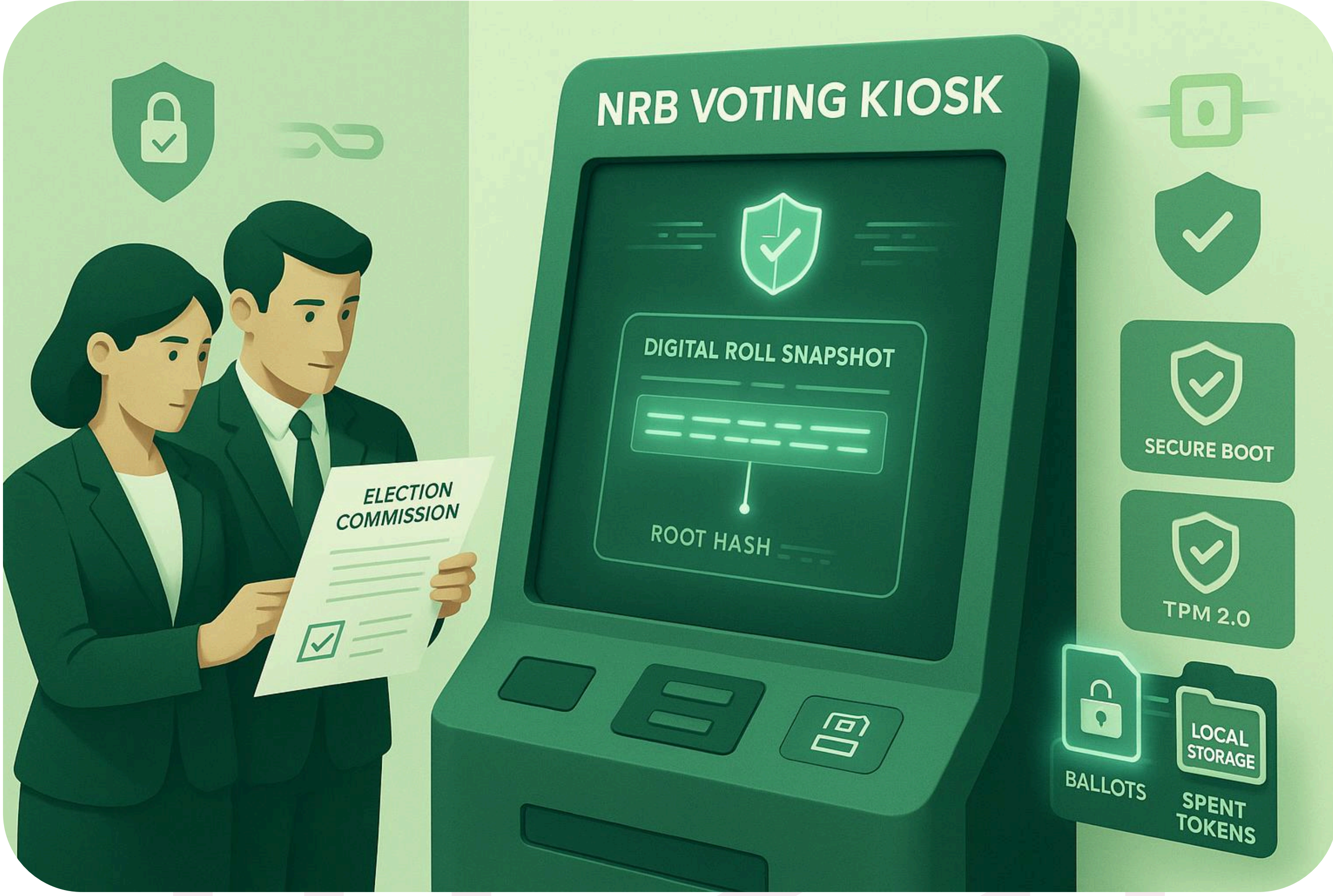
Merkle tree generated → root hash published:

On BD-VoteNet blockchain

In newspapers & embassy notice boards

This ensures any change is instantly detectable

Offline Kiosk Setup



Kiosks equipped with:

- Roll snapshot (matching published root)
- Secure Boot + TPM 2.0 + measured boot
- Encrypted local storage for ballots & spent tokens
- Observers verify snapshot ID matches EC record before polls open

Encrypted Hard Drive Delivery

Roll snapshot, election parameters, and validator keys stored on sealed, encrypted drives

Delivered physically to each embassy and offline site

Chain-of-custody log signed by EC & observers

Poll Opening

Kiosk boots → self-attestation check passes

Prints Opening Manifest:

Device ID

Snapshot ID

Empty spent-token set hash

Observers countersign and keep copy

Voter Authentication (Offline)

Voter presents BD PASS QR + fingerprint/face scan

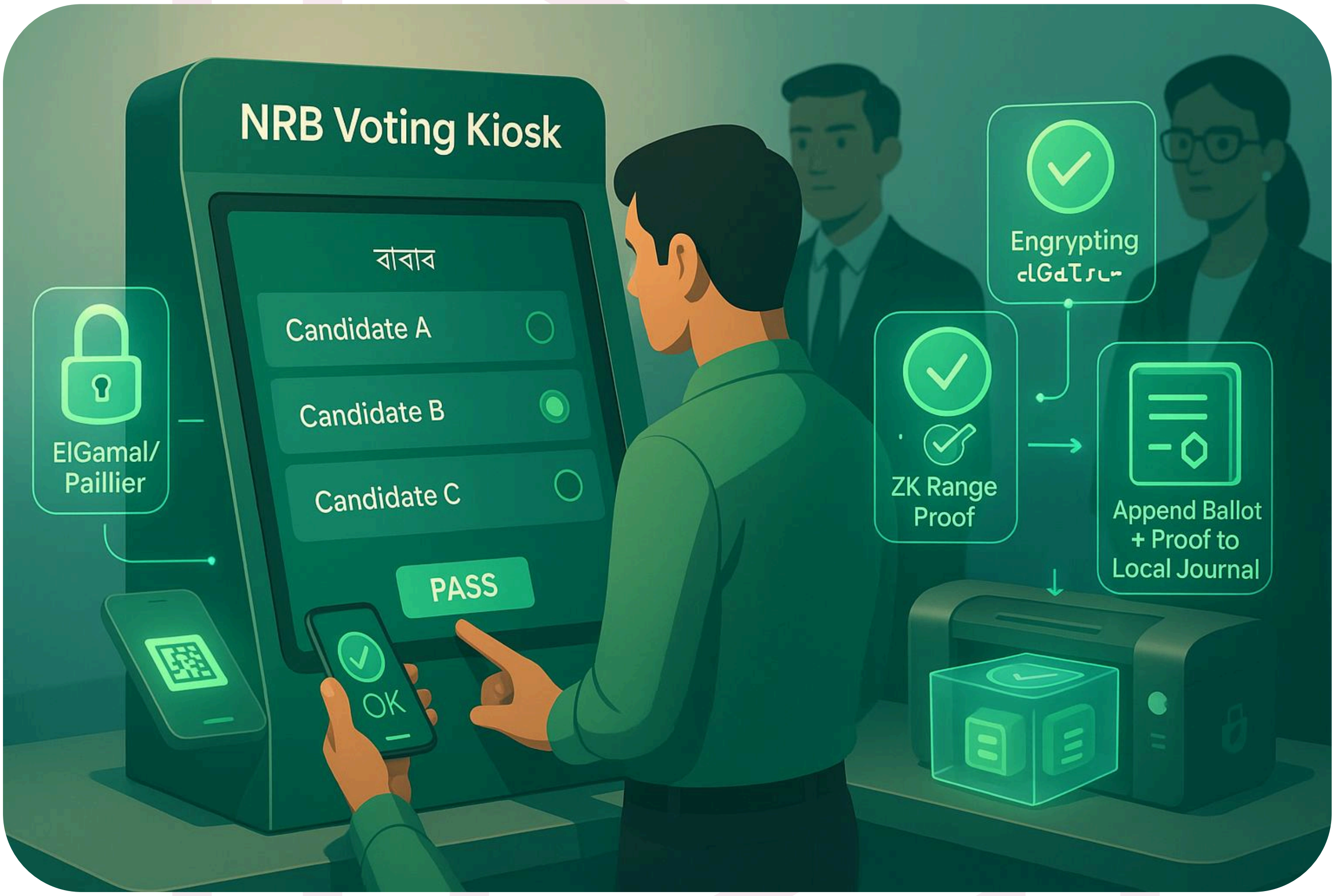
Kiosk verifies ZK membership proof against local roll snapshot

Checks local spent-token set:

If unused → proceed

If already used → reject & print rejection slip

Casting the Vote



Voter selects candidate on screen (Bangla/English/Arabic UI)

Kiosk:

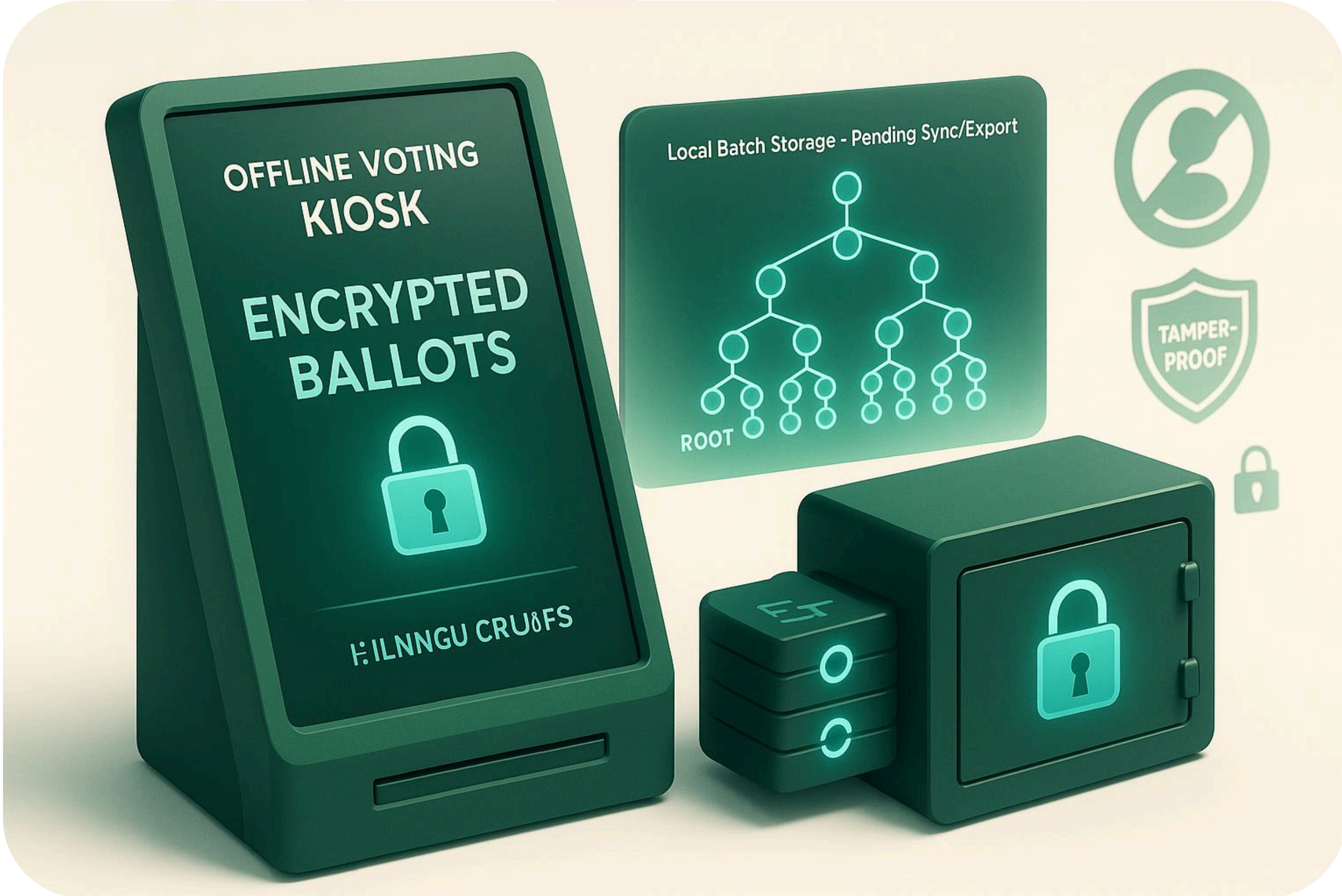
Encrypts vote (ElGamal/Paillier)

Generates ZK range proof (valid choice)

Adds token to local spent set

Appends ballot + proof to sealed local journal

Local Batch Storage



Votes remain stored locally until sync or export

Kiosk updates local Merkle accumulator for each new vote

No ballot decryption possible on kiosk

Poll Closing

At close time:

- Kiosk stops accepting votes

Prints Final Batch Manifest:

- Batch Merkle root
- Total votes
- TPM counter value
- Device signature
- Observers countersign
- Export sealed vote batch to WORM media (SD card / LTO tape)
- Place in tamper-evident bag with signed custody log

Transport to EC / Relay Node

Sealed media delivered physically to EC ingest station or relay node

Chain-of-custody verified upon arrival

Import & Verification

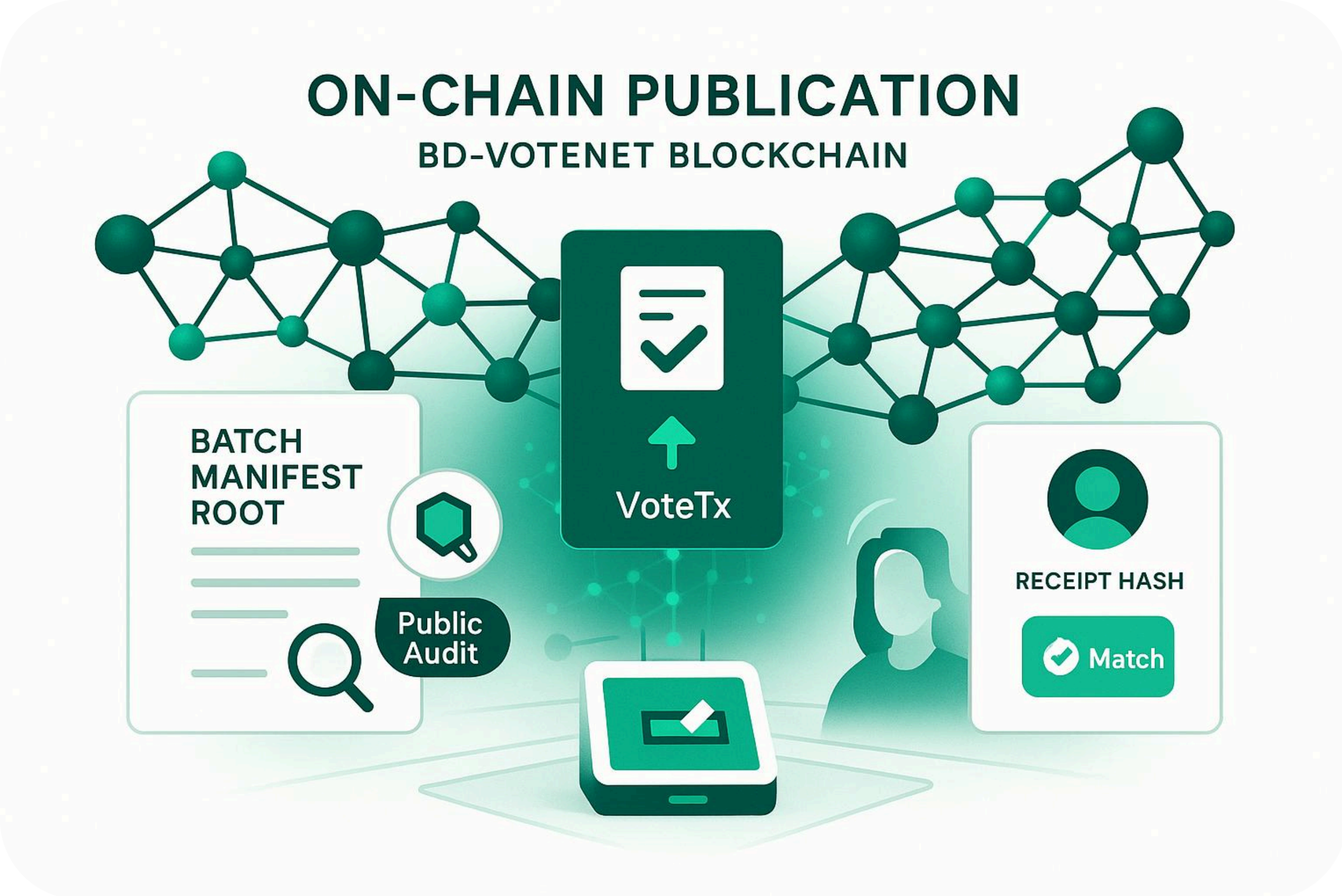
EC import tool:

- Checks device attestation & manifest hash
- Verifies all ZK proofs
- Compares batch root to printed manifest

Detects duplicate tokens:

- If duplicate → earliest TPM-counter timestamp wins
- Later one rejected & logged publicly

On-Chain Publication



Valid ballots posted to BD-VoteNet blockchain as VoteTx

Batch manifest root also published for public audit

Anyone can verify their receipt hash exists in final batch

Homomorphic Tallying & Proof

All encrypted ballots summed homomorphically

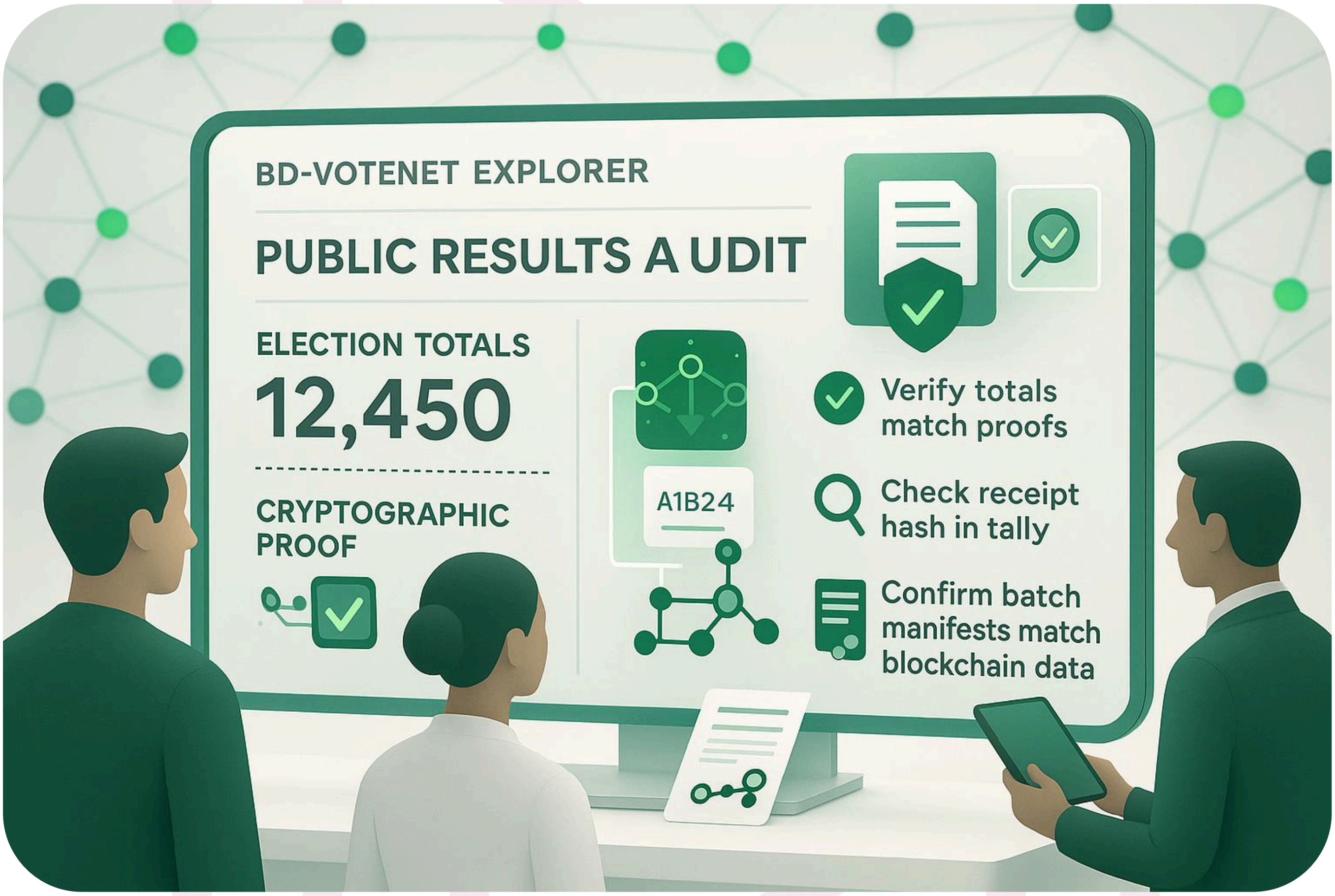
Validators decrypt totals collaboratively (threshold decryption)

Groth16 proof generated to show:

All counted votes were valid & included

No extra or missing ballots

Public Results & Audit



EC publishes results + proofs on BD-VoteNet Explorer

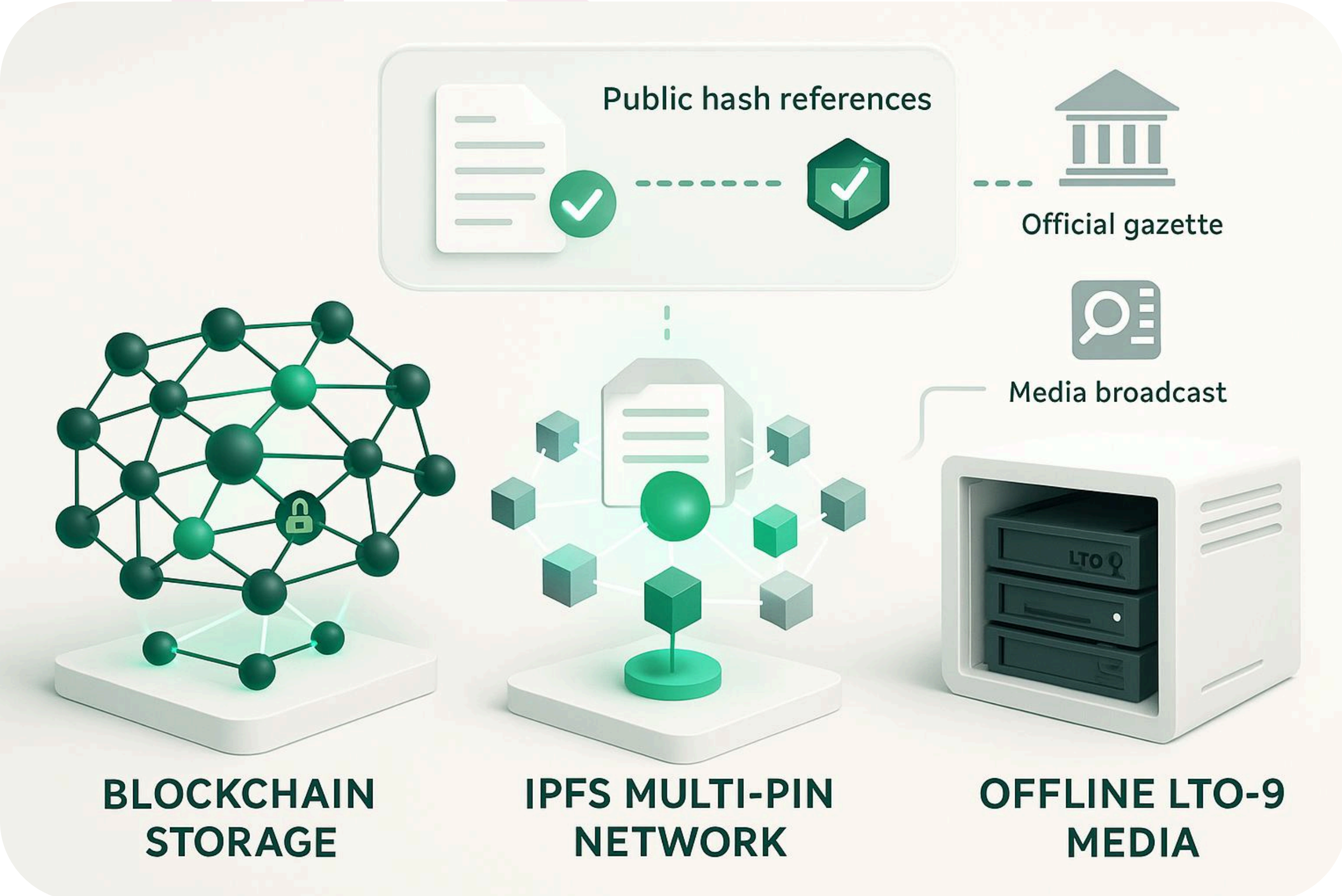
Public & observers can:

Verify totals match proofs

Check receipt hash in tally

Confirm batch manifests match published blockchain data

Archival



Final election data stored:

- On blockchain (permanent)
- On IPFS multi-pin network (distributed copies)
- On offline LTO-9 media for long-term storage
- Public hash references published in official gazette & media